23CSE452
BUSINESS ANALYTICS

Career Market Intelligence Engine: Salary Prediction and Skill Demand Analytics for Indian Job Postings

Group 3 (Project Report)

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Declaration and acknowledgement

We declare that this report describes work done by our team for the course named on the cover page. The analysis, code, figures and dashboard were produced by the team members listed in Section 8.2. Data, ideas, figures and code that come from other sources are cited in the text and listed in the references. The report has not been submitted for any other course or degree.

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Abstract

Job boards show what employers ask for, and sometimes what they pay, but few studies turn this into numbers that a job seeker, a recruiter or a training institute can act on in India. We collected 14,085 full-text postings from the Naukri job board. Of these, 10,780 map to one of ten business and technology roles, and 1,358 state a usable yearly salary range. We estimate pay for employers the model has not seen, and we return an 80% range whose coverage we check by role. We then test which skills go with higher pay, whether a description identifies its role, whether it predicts that the employer states pay, and which skills would move a posting toward another role. Google Trends data for 20 skills shows how search interest differs from employer demand. A tuned gradient-boosting model reaches R^2 0.767 and a mean error of ₹3.8 lakh on 257 unseen-employer postings (role median: 0.335 and ₹5.6 lakh), and its 80% range held 83.3% of stated salaries. Experience carries most of the signal. Of 170 skills, 17 show a pay association that survives false-discovery control, and their ranking is unstable across employer halves. Only 12.9% of postings state pay, and the description predicts that with ROC-AUC 0.87. The work is delivered as four notebooks and a five-section dashboard.

Keywords: job postings, salary estimation, conformal prediction, text mining, time series, pay transparency, India.

Executive summary

Problem. A person deciding whether an offer is fair, a recruiter setting a salary band and a trainer choosing what to teach all rely on job postings, yet about nine in ten postings on the board we studied do not state a salary. Where a salary is stated it is a range, and skills lists mix useful tools with filler.

What we did. We studied 14,085 postings for ten roles. We built a salary estimate with an 80% range, tested which skills are linked to pay, checked whether a description identifies the role, and checked who states pay at all. We also compared employer demand for 20 skills with five years of Google search interest. All results are available in an interactive dashboard.

Main evidence. For employers the model had not seen, the typical error was ₹3.8 lakh against ₹5.6 lakh for guessing the role median, and 60% of estimates were within 30% of the stated salary. The 80% range held for 83% of postings, but only 59% for Software Engineer and 64% for Data Scientist. Experience is the strongest input. Only 17 of 170 skills show a clear pay association, and Python and SQL, the two most requested skills, show none. Only 12.9% of postings state pay, mostly from the same few employers. Search interest for Python and SQL fell by about a third and a half over five years while a quarter of postings still ask for them.

Value and actions. A job seeker can compare an offer with a range and similar postings, a recruiter can benchmark a role with a stated reliability, and a trainer gets a short list of skills that separate neighbouring roles. Negotiate from the range, treat Python and SQL as entry requirements, report how many postings stand behind any benchmark, and collect postings again over time before relying on skill pay figures.

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List of abbreviations

CV: cross-validation
DiD: difference-in-differences
DML: double machine learning
EDA: exploratory data analysis
FDR: false-discovery rate
KPI: key performance indicator
LDA: latent Dirichlet allocation
MAE: mean absolute error
MASE: mean absolute scaled error
MAPE: mean absolute percentage error
NMF: non-negative matrix factorisation
NMI: normalised mutual information
ROC-AUC: area under the receiver operating characteristic curve
STL: seasonal and trend decomposition using loess
SVD: singular value decomposition
TF-IDF: term frequency, inverse document frequency

1. Introduction, Business Objectives and Literature Review

1.1 Context and scope

Many Indian job seekers use Naukri to compare roles, and recruiters use it to see what rivals ask for. The board therefore holds a large record of requested skills and, for some postings, salaries. Three groups use this record. Job seekers and students want to know what an offer should pay and which skills to learn. Recruiters and human resources teams want a market benchmark for a role. Training institutes want to teach skills that employers ask for and that pay. The business problem is that most postings state no salary, the stated salary is a wide range, and skills lists do not say which skills matter for pay.

The scope is ten roles (Software Engineer, Data Analyst, Data Scientist, Data Engineer, DevOps and Cloud Engineer, Business Analyst, Product Manager, Project Manager, Digital Marketing and Graphic Designer) on one job board, with the salaries stated yearly in rupees. Search interest comes from Google Trends for India. The overall objective is to turn postings into decision support for the three groups above. The analytical questions and their measurable targets are in Table 1.

Table 1. Business objectives, KPIs and success criteria.

Business objective	KPI / unit	Baseline / target	Relevant unit or method
O1. Estimate pay for employers not seen in training	R^2 on log salary; mean error in ₹ lakh	Baseline: role median (R^2 0.335, ₹5.6 lakh). Target: clearly better on both	Unit 2
O2. Give a range that holds	Share of stated salaries inside the 80% range (%)	Target 80%, with coverage reported by role	Unit 2
O3. Find skills that go with pay	Skills with a pay association after false-discovery control (count of 170)	Baseline: skills found by chance at $p < 0.05$ (about 8.5 of 170)	Unit 3, text mining
O4. Tell whether a description identifies its role	Accuracy and macro-F1 on unseen companies	Baseline: largest role, 19.8% accuracy	Unit 3, text mining
O5. Find out who states pay	F1 and ROC-AUC for stating pay	Baseline: always "does not state" (F1 0, AUC 0.5)	Unit 3, text mining
O6. Compare employer demand with search interest	Skills rising or declining; forecast error (MASE)	Baseline: naive forecast (MASE 1.035)	Unit 3, time series

Figure 1 shows how the units connect. Raw postings are cleaned once in Section 2. The cleaned data feeds the exploratory analysis (Unit 1), the salary model (Unit 2) and the two Unit 3 methods, text mining and time series. Their outputs are joined in Section 6 and shown together in the dashboard in Section 7.

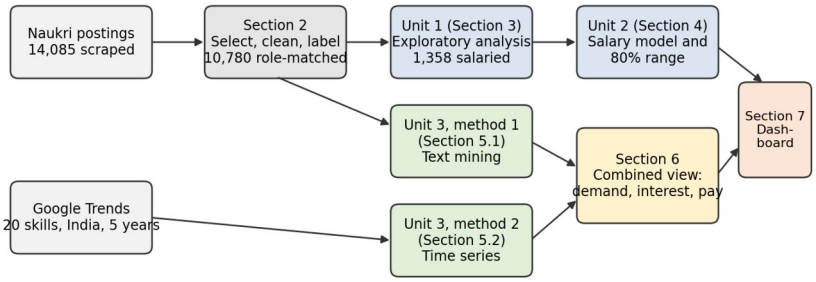


Figure 1. End-to-end workflow connecting the units and the application. Tool: Python (Matplotlib). Source: figure drawn for this report.

1.2 Related analytics work and research gap

Bana trained a language model on the text of postings from a US aggregator and reported that the words explain a large share of salary variation, much more than titles and locations alone [2]. The summary we read does not define the metric, so we use it as context, not as a number to match. Deming and Kahn linked skill requirements in US postings to wages within occupations and firms [10]. Ao and co-authors compared ways of finding skills in a UK set of over one million advertisements by how much wage variation each explains, and found that topic modelling explained more than keyword dictionaries [1].

Batra, Michaud and Mongey studied how little wage information US online postings carry: about 14% contain any wage, the range averages about 28% of its midpoint, and high-wage firms disclose less [3]. Hazell and Taska used posted wages to study how rigid pay is for new hires [13]. For India we found a framework for mining skills from computer-science advertisements that reports no quantitative results [24].

These studies leave three needs unmet for our setting. First, none gives a salary estimate with a stated range checked on employers not used in training, for an Indian board. Second, the pay-transparency studies describe who discloses but, in the sources we read, do not ask whether the wording of a posting predicts disclosure. Third, skill-and-wage studies report coefficients without checking how stable they are on a second set of employers. Regression with employer-grouped validation, conformal ranges [22], and a stability test address these. The numerical comparison with these studies is in Section 6.2.

2. Dataset Collection, Validation and Preparation

2.1 Provenance, coverage and challenges

The postings come from the Naukri job board [18], collected with a commercial scraper (an Apify actor named thirdwatch/naukri-jobs-scraper) in a single snapshot. Each posting has a title, company, locations, work mode, experience range, employer-tagged skills, industry, the full description, and sometimes a salary range. Search interest comes from the Google Trends service [11], queried through the pytrends package for 20 skills in India over five years of weekly data, in batches of five terms with a shared anchor term (SQL) so that batches can be placed on one scale. No synthetic data was used.

The scrape returned 14,085 postings and 63 columns, of which we kept 28. The unit of observation is one posting. Table 2 shows the counts through preparation and Table 3 a sample of the final salary table.

Table 2. Record counts through preparation.

Step	Postings	Note
Scraped	14,085	63 columns, unique job id
Mapped to one of ten roles from the title	11,589	2,401 matched no role, 63 matched two, 32 hit an exclusion rule
State a salary (role-matched, salary notebook)	1,457	12.6% of role-matched postings
Final salary table (Units 1 and 2)	1,358	1,017 employers; screens in Section 2.3
Text-mining corpus, distinct descriptions (Unit 3)	10,780	Descriptions of at least 50 characters, identical ones kept once
State a salary in the text-mining corpus	1,389	12.9%; 1,348 after screens, from 1,016 employers

Table 3. Sample of the final salary table (title and company omitted).

Role	City	Work mode	Years asked	Skills tagged (first three)	Stated range (₹ lakh)	Split
Software Engineer	Pune	Hybrid	5-10	microsoft fabric, azure data factory, databricks unified data analytics platform	15-27.5	train
Data Scientist	Hyderabad	Work from office	6-11	machine learning, large language model, python	13-20	train
Data Engineer	Mumbai	Hybrid	5-10	snowflake, sql, data build tool	15-30	train

Collection had four practical limits. Runs were merged and repeated job ids removed. About 88% of postings carry no salary, which sets the size of every pay result. Some ranges are open-ended (starting at 0) or look monthly. Search interest is an index scaled to the largest term in

each request, so batches were rescaled with the shared anchor. The collection was a single snapshot, so the results describe one point in time.

2.2 Data quality, privacy and ethics

Quality checks. Missing salary is the main gap (87.9% of raw postings). We also found repeated descriptions (790 in the role-matched set), repeated title-company-range rows (9), open-ended ranges starting at 0 (dropped) and midpoints under ₹100,000 a year that look monthly (32 dropped). Role sizes are uneven: Digital Marketing has 370 salaried postings and Data Scientist 72.

Privacy and bias. The data is public advertising text. Recruiter names, emails, links and logos were dropped at the first step, so no personal data about individuals is analysed. Company names appear only in local dashboard files, and the raw postings are kept locally and not redistributed, in line with the data minimisation expected under India's Digital Personal Data Protection Act, 2023 [12]. Only about one posting in eight states pay, mostly from a minority of employers, so pay results describe postings that disclose.

2.3 Transformations and usable dataset

Descriptions were lower-cased, links and addresses removed and whitespace collapsed, with no stemming. Seniority came from title keywords, city from the first listed location grouped into hubs, and experience from the midpoint of the stated range. Skills are the 200 most common employer tags; a posting asks for a skill if it was tagged or named in the text. The salary target is the log of the midpoint of the stated range. Model-specific steps such as the vocabulary, SVD and encoders were fitted on training postings only, and every split keeps each employer in one part (1,101 training and 257 test postings).

Table 4. Mapping of the final fields to the units and methods.

Variable	Type / meaning	Origin / preparation	Used in
role	Category, 10 levels	Title rules, one role per posting	U1, U2, U3 M1
experience (min, max, midpoint)	Years, number	Parsed from the stated range	U1, U2, U3 M1
seniority	Entry, Mid, Senior	Title keywords	U1, U2, U3 M1
city, work mode, industry	Category	First listed city grouped to hubs; board fields	U1, U2, U3 M1
skill tags	List of text	Employer tags, 200 most common used	U2, U3 M1
description	Text	Cleaned; TF-IDF, 50-component SVD, topics	U2, U3 M1
salary min, max	Rupees per year	Stated yearly range; screens in Section 2.2	U1, U2

Variable	Type / meaning	Origin / preparation	Used in
states pay	Yes or no	Salary field holds a yearly range	U3 M1
weekly search interest	Index 0-100	Google Trends, rescaled with SQL anchor	U3 M2

3. Exploratory Data Analysis and Visualization (Unit 1)

The questions behind this section are: how is pay spread, what separates high from low pay, who states a salary, and is there enough text to learn from? All figures come from the notebook career_market_intelligence.ipynb (Python, Matplotlib and Seaborn) [14].

Salary is strongly right-skewed (skewness 1.72) and close to symmetric on the log scale (−0.04), so the models predict log salary. The median midpoint is ₹9.0 lakh against a mean of ₹12.3 lakh, and the middle 90% run from about ₹2.4 to ₹32.5 lakh. The log histogram has two humps, near ₹4 to 5 lakh and ₹20 lakh, which correspond to the lower-paid marketing and design roles and the technical and management roles. In the right panel the stated maximum is typically 1.67 times the minimum, so a single number hides a wide range.

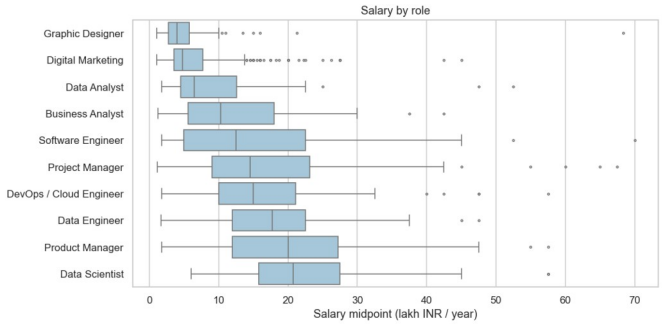


Figure 2. Salary midpoint by role. Boxes show the middle half; whiskers and dots show spread and outliers. Source: career_market_intelligence.ipynb, cell eda_role.

Role separates pay more than any other field in the table. Median pay is ₹4.0 lakh for Graphic Designers and ₹4.75 lakh for Digital Marketing, against ₹20.8 lakh for Data Scientists and ₹20.0 lakh for Product Managers, about a five-fold spread. The boxes overlap, so role alone cannot give a salary. Experience is the strongest single signal: the correlation of log salary with the experience midpoint is 0.71. Median pay rises from ₹3.4 lakh (Entry) to ₹6.2 lakh (Mid) and ₹13.0 lakh (Senior).

About one posting in eight states a yearly range (12.6% of role-matched postings). The rate runs from 5.5% for Software Engineer to 18.7% for Digital Marketing. On experience and company rating, postings that state pay look like those that do not (mean experience 6.1 against 6.0 years,

rating 3.55 for both), which is reassuring but incomplete. The 1,358 postings come from 1,017 employers; the largest has 16 postings (1.2%) and 84% appear once, so we group all validation by employer.

4. Feature Engineering and Predictive Modelling (Unit 2)

Target and predictors. The target is the log of the midpoint of the stated yearly range. Predictors are experience (midpoint, square and span), role, seniority, city group, work mode, industry, company attributes, education, the 60 most common skill tags and a 50-component truncated SVD of the TF-IDF description. Text is reduced to 50 components so that 1,101 training rows are not swamped by thousands of word columns.

Protocol and leakage audit. The 257 test postings (204 companies) were set aside before any model was fitted and used once. Models are compared by five-fold cross-validation grouped by company on the 1,101 training postings. Encoders, the vocabulary and the SVD are fitted inside each training fold. We checked that no employer is in both parts, that salary amounts quoted in the text are not driving the result (removing pay amounts from the text changed R^2 by 0.002), and that the target does not enter any feature. A randomised search over XGBoost settings (20 draws) used the training postings only [7].

Table 5. Models on the same folds and the held-out test.

Model	Validation scheme	Test metrics	Interpretability / bias evidence
Global median (baseline)	5-fold CV by company	CV R^2 −0.007; mean error ₹7.8 lakh	No inputs; sets the floor
Role median (baseline)	5-fold CV by company; test	CV R^2 0.372; test R^2 0.335, error ₹5.6 lakh, 36.6% within 30%	Shows how little role alone explains
Ridge	5-fold CV by company	CV R^2 0.671; error ₹4.6 lakh	Coefficients; close to boosted models
Random forest	5-fold CV by company	CV R^2 0.671; error ₹4.5 lakh	Importance by group
Stacking (Ridge, forest, XGBoost)	5-fold CV by company	CV R^2 0.704; error ₹4.2 lakh	Hard to explain; tied with XGBoost
XGBoost, tuned (selected)	5-fold CV; held-out test	CV R^2 0.696; test R^2 0.767, error ₹3.8 lakh, 59.9% within 30%	Permutation importance, error by role

Stacking and XGBoost differ by 0.009 in R^2, far inside the fold-to-fold spread of about 0.05, so neither is a clear winner. Ridge is only 0.03 behind, which says the signal is mostly additive. We selected XGBoost because it is a single model, nearly as accurate as the stack, and supports the explanations below. Tuning did not help (R^2 0.695 to 0.696), so the limit is the information in the postings, not the settings.

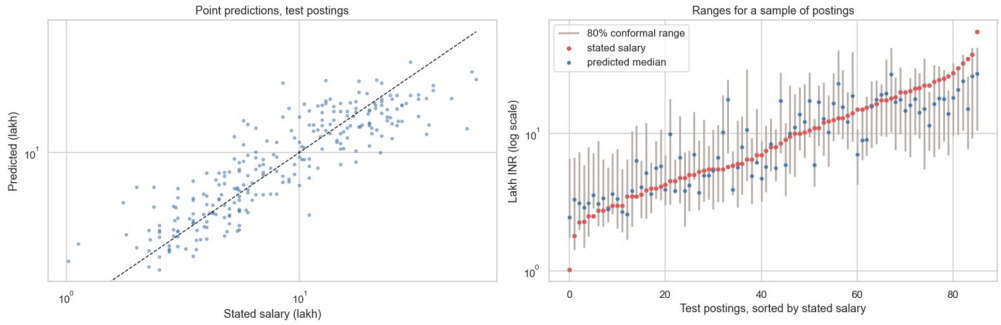


Figure 3. Predicted against stated salary on the 257 test postings (left) and 80% ranges for a sample, sorted by stated salary (right). Source: career_market_intelligence.ipynb, cell eval_plot.

On unseen employers the tuned model reaches test R^2 0.767 and a mean error of ₹3.8 lakh, against 0.335 and ₹5.6 lakh for the role median, and 59.9% of estimates fall within 30% of the stated salary (36.6% for the role median). The 80% range uses conformalized quantile regression [22]: quantile models at the 10th and 90th percentiles are fitted on a proper training part, and a calibration part, separate by employer, widens them. The range held 83.3% of test salaries, against 70.4% for the uncalibrated bands.

What the model uses. Removing experience lowers cross-validated R^2 by 0.109 and removing the description by 0.023; removing any other group costs 0.006 or less. Shuffling a group on the test postings gives the same order (experience 0.61, description 0.23). This does not mean role and seniority carry no information. Each alone explains much of the variation (role 40%, experience 51%), but the description names the role with about 90% accuracy and seniority moves with experience, so once those inputs are present the others add little.

Subgroup check. The overall figure hides differences. Coverage is 58.8% for Software Engineer (17 test postings), 63.6% for Data Scientist (11), 72.2% for Product Manager (18) and 75.0% for Business Analyst (16), against 96.9% for Graphic Designer (32). By experience it is 76.1% for 5 to 8 years and 78.7% for 8 to 12 years. The guarantee of conformal prediction holds over all postings together, not within each role, and the small roles have too few test postings to say more. We report this as a limit and recommend that users of the range read it with the role-level figure. For business use, the model beats the role median clearly for every role in typical error, so it meets O1, and it meets O2 overall but not for the four roles above.

5. Advanced Analytics (Unit 3): Two Selected Methods

We chose text mining and time-series analysis. Text mining suits the problem because pay, role and disclosure all depend on what a posting says. Time series suits the question of how skill demand compares with public interest over time. Both are presented with the same structure.

5.1 Method 1: Text mining

Business question and inputs. Which skills go with higher pay, can a description identify its role, who states pay, and which skills would move a posting toward another role? The input is the 10,780 role-matched postings with distinct descriptions; the pay questions use the 1,348 that state a salary after screens. The notebook is text_mining.ipynb.

Algorithms and settings. Skills were found from employer tags and whole-word matching in the text. Topics came from LDA and NMF (K = 8, chosen by held-out perplexity) [5]. Posting clusters used K-means. Skill bundles came from Louvain community detection on a co-occurrence network built from lift [6]. Pay associations were estimated three ways for 170 skills with at least 25 salaried postings: least squares with company-clustered errors, a joint ridge regression with company bootstrap, and double machine learning with five-fold cross-fitting [8], using role, experience, seniority, city, work mode and industry as controls, and Benjamini-Hochberg false-discovery control at 10% [4]. Role fit compared ten models under company-grouped cross-validation. Disclosure classification compared seven models, with salary amounts removed from the text, and ranked words by weighted log-odds [17]. The career switch search (actionable recourse) used a multinomial logistic regression on the 200 skill indicators and added, one at a time, the skill that most raised the target role's lead, limited to skills at least 10% of the target role's postings ask for [25].

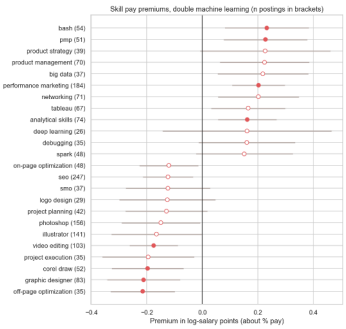


Figure 4. Skill pay associations from double machine learning, with 95% intervals. Filled points survive false-discovery control. Source: text_mining.ipynb, cell forest.

Pay associations. Of 170 skills, 32 have $p < 0.05$ by least squares, about 8.5 expected by chance, and 15 survive correction; double machine learning gives 31 and 8, and 17 skills are clear in at least one estimator. The largest positive double-ML values are bash (+26%), product management (+25%), networking (+23%), performance marketing (+23%) and analytical skills (+18%); the largest negative are off-page optimisation (−19%), graphic designer (−19%), CorelDraw (−18%) and video editing (−16%). Python (+8%) and SQL (+4%) are not distinguishable from zero. However, estimating on a random 80% of companies and again on the rest gives a rank agreement of only about 0.28, so exact percentages are unreliable. Adding skills to the salary model lifted held-out R^2 by 0.01 or less (Section 4).

Role and disclosure classification. On 2,064 test postings from unseen companies a linear model reads the role from the description with accuracy 0.897 and macro-F1 0.881 (largest-role guess: 0.198 and 0.033). Removing the title's own words leaves it at 0.896 and 0.881, so the model is not copying the title; reading only the first 500 characters lowers cross-validated macro-F1 to 0.839. Skills alone place 90.3% of postings in the right role. Only 12.9% of postings state pay, so always answering "does not state" is 87.1% accurate and useless. Table 6 gives the models.

Table 6. Predicting whether a posting states pay (five-fold cross-validation by company, 10,780 postings).

Model for "states pay"	ROC-AUC	Precision	Recall	F1
Logistic regression, words and attributes	0.882	0.415	0.688	0.518
Logistic regression, words	0.871	0.440	0.629	0.518
Linear SVM, words	0.863	0.435	0.598	0.504
MLP on 100 SVD components	0.851	0.404	0.623	0.490
Complement naive Bayes, words	0.839	0.399	0.546	0.461
XGBoost, posting attributes	0.819	0.351	0.575	0.435
Logistic regression, role only	0.593	0.157	0.649	0.253
Always "does not state"	0.500	0	0	0

The words predict disclosure well (F1 0.52, and 0.552 with ROC-AUC 0.871 on 2,064 held-out postings), and the posting attributes add almost nothing. Compressing the words by SVD first does not help. Reading only 500 characters lowers F1 to 0.42, because the cues sit in the closing boilerplate. The pattern is mostly an employer habit: 76.5% of postings come from companies that never state a salary, and 85.5% of the postings that do come from companies that state one in at least half of their postings. Inside a role, employers that stay silent point to 3.6% lower pay, with an interval from −7.2% to +2.4%, which includes zero.

Career switch. Across all 90 role moves, a posting reaches the target role within five added skills in 48.1% of cases and within ten in 84.3%, with a median of five. The baseline that adds the target role's most common missing skills reaches 29.2% and 61.7%. A flip only means the classifier reads the posting as the target role.

5.2 Method 2: Time-series analysis

Business question and inputs. Which skills are people searching for more or less, and do skills that employers still ask for show falling interest? The input is weekly Google Trends interest for India for 20 skills over five years (261 weeks), rescaled with the SQL anchor. The notebook is method2_time_series.ipynb. Anchor agreement across batches was high (minimum correlation 0.999), and a repeat pull with plain-word queries gave the same class for 13 of 20 skills.

Algorithms and settings. Weekly series were smoothed with an eight-week average. Each skill is labelled rising, stable or declining from the five-year change (cut-off 10%), and the label is confirmed with a Mann-Kendall trend test [16]. Forecasts used a naive baseline, exponential smoothing, ARIMA and Holt-Winters, scored by rolling-origin backtests with MAPE and MASE [15], with conformal intervals. Seasonality used STL decomposition [9], computed with statsmodels [23]. To test whether code skills lost interest after AI coding assistants appeared, five code skills were compared with seven tool and platform skills around November 2022 by interrupted time series and by a difference-in-differences whose p-value comes from all 792 ways of choosing five treated skills out of twelve.

Results. Of 20 skills, 3 are rising (Deep Learning, Google Cloud, Machine Learning), 5 stable and 12 declining, and six have trends too small to judge. Python fell 34% and SQL 56% in five years. No forecasting method beats the naive forecast on 98 rolling windows (median MASE 1.035 naive, 1.076 smoothing, 1.126 ARIMA, 1.115 Holt-Winters), so forecasts are given with intervals and not as point claims.

After the release, code skills lost about 0.32 log points (roughly 27%) more search interest than the control skills, and only 3 of 792 groupings give a gap that low or lower ($p = 0.005$). No skill drops sharply at the release. Instead the code skills were rising before and bend downward afterward, and the gap is largest in 2025, so the date is wrong for the release itself. The test cannot separate AI assistants from other causes. **Limits:** Google Trends is a sample and

measures attention, not hiring; the original pulls were not kept, so validation was repeated on fresh pulls.

6. Comparative Analysis and Business Impact

6.1 Comparison of the two Unit 3 methods

Text mining works on what postings say (10,780 descriptions, 1,348 with salary) and gives pay associations, classifiers and skill moves; time series works on 20 weekly search series and gives trend labels, forecasts with intervals and a break test. The methods complement each other. Python and SQL are asked for in about a quarter of postings each, while their search interest fell 34% and 56%. This fits skills that have become assumed knowledge, and neither carries a clear pay premium. Machine Learning is the only skill high on both demand (12.6% of postings) and rising interest (+40%), with a pay association of +21% in the 20-skill analysis and +14% in the 170-skill analysis, where it is not clear after correction. Figure 5 shows the combined view.

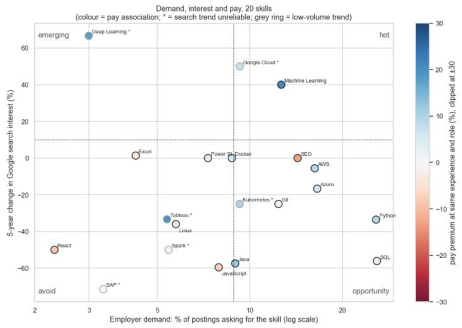


Figure 5. Employer demand against five-year change in search interest for 20 skills, coloured by pay association. Source: combined_insight.ipynb, cell plot.

By the fixed quadrant rule, one skill is hot, seven are in the opportunity quadrant (high demand, flat or falling interest), six are avoid and six have a trend too small to place.

6.2 Comparison with existing studies

Table 7. Current findings against existing studies.

Criterion	Batra et al. [3]	Ao et al. [1]	Bana [2]	Current project
Data and setting	US, all online job posts	UK, over 1 million advertisements	US aggregator, about 1 million postings	India, one board, 14,085 postings, 1,358 with salary
Methods and validation	Descriptive statistics of disclosure	Wage regressions; skill methods judged by variation explained	Fine-tuned language model; 80/20 split	Boosted model, employer-grouped CV; conformal range; stability tests
Comparable findings	About 14% state any wage; range 28% of midpoint; firm concentration	Dictionaries about 20%, topic model about 45% of wage variation	Text explains "87%" (metric not defined in the source we read)	12.9% state pay; range 50% of midpoint; 76.5% of postings from never-disclosers; R^2 0.767 test
Limits	No study of wording	Different metric from ours	Not comparable to our R^2	Small test set; one snapshot

Our disclosure share (12.9%) is close to the US 14%, and disclosure is concentrated in few employers in both. Our stated ranges are wider: the median top-to-bottom ratio is 1.67, a span of 50% of the midpoint, against 28% in the US data. Like Ao et al., we find that text measures capture more than a skill list: topic shares alone explain R^2 0.36 of log salary and the 50-component text summary 0.52, while adding skills to experience and role lifts R^2 by about 0.01. A numerical comparison with the language-model result is not valid, because its metric is undefined in the summary we read and the data differ in size by about three orders of magnitude. The project adds a role-level check of the salary range for an Indian board, a disclosure classifier and a stability test of skill premiums. It does not claim better accuracy than the other studies.

6.3 KPI-based business impact and scenarios

Measured outcomes. For the 257 unseen-employer postings the model's mean error was ₹3.77 lakh against ₹5.60 lakh for the role median, which is ₹1.83 lakh (33%) lower per posting, and the 80% range held 83.3% of the time. These are measured on held-out data.

Table 8. Sensitivity of range reliability by role.

Scenario	Assumption	Coverage of the 80% range	Reading
Best	Roles like Graphic Designer (32 test postings)	96.9%	Range is wider than needed
Base	All roles together (257 postings)	83.3% (95% interval 78.2% to 87.3%)	Close to target
Worst	Roles like Software Engineer (17 postings)	58.8%	Range too narrow; widen it by hand

7. Interactive Application, Figure Index and Reproducibility

7.1 Dashboard

The dashboard is a Next.js application [19] with Observable Plot charts [20]. It runs locally from the repository folder named frontend, with these commands: `cd frontend`, `npm install`, `npm run dev`, then open http://localhost:3100. A hosted copy is at https://career-market-intelligence.vercel.app. A production build uses `npm run build` and `npm start`. Notebooks write JSON files with the final aggregates (for example salary.json, textmining.json, trends.json, gap.json, dataset.json, switch.json), and the application reads them, so the pages and the notebooks cannot disagree.

A top bar has five sections and a filter strip (role, city, experience, work mode) that carries across pages. **Overview** shows KPIs and pay by role, experience and city for the selection. **Pay** has the

estimator with its 80% range, an offer check and range reliability. **Skills** has the demand, interest and pay map, pay associations and a skill planner. **Careers** has the career switch planner. **Data and method** has the dataset, who states pay and the method notes. KPIs, pay charts, the dataset table, the offer check and the career switch search are computed in the browser from the filter; model comparisons, premiums, topics and trends come from the notebooks and are marked as not filtered.

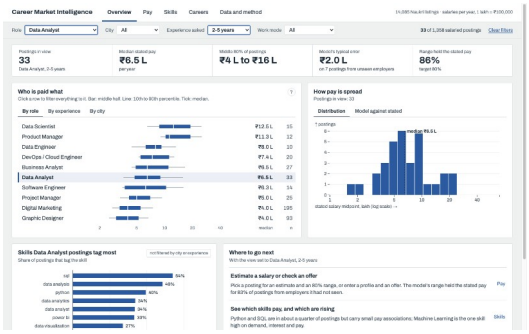


Figure 6. Dashboard Overview filtered to Data Analyst postings asking for 2 to 5 years: 33 postings, median ₹6.5 lakh. Source: frontend application, route /?role=Data Analyst&band=2-5 years.

Tests and limitations. A scripted browser session of about 40 steps (filters, tabs, estimator, offer check, skill lookup, career switch, dataset search) ran with no console errors, and the build and type check pass. Phone layout was not tested on a device. The dashboard files contain scraped titles and company names, so the application is for local use only.

7.2 Figure index and reproduction map

Every figure comes from the notebook named in its caption, in the cell named there. Appendix A lists all figures with their sections. The traceability matrix in Section 8.1 links each objective to its figure or file.

7.3 Code and environment

The repository is https://github.com/anusanth26/Career-Market-Intelligence. Folders are data, notebooks, scripts, reports, frontend and config, and the README files describe the tree. Dependencies are in requirements.txt (pandas, scikit-learn, xgboost, statsmodels, networkx, pytrends and others) [21], with Python 3.10 and Node.js 22. To reproduce, install the requirements, run the notebooks in the order career_market_intelligence, text_mining, method2_time_series, combined_insight (each writes the JSON the dashboard reads), then start the dashboard. No keys are stored in the repository. The raw postings and Trends pulls are not

distributed because the board's terms may restrict copying. Seeds are fixed at 42 and splits group by employer.

8. End-to-End Validation, Traceability and Contributions

8.1 Completion and validity

Unit 1 is covered by Section 3 (descriptive statistics, distributions, segment comparisons and the disclosure anomaly), Unit 2 by Section 4 (baselines, four models, leakage audit, explanation and subgroup checks), Unit 3 by Section 5 (text mining and time series), and the application by Section 7 (screenshots and a scripted test). Notebooks execute in order with no errors.

Threats to validity. One board and one snapshot limit generalisation and rule out claims about change over time. Only 12.9% of postings state pay, so pay results describe disclosing postings. Role labels come from title rules. Test sets are small (257 postings, 11 to 33 per role). Skill premiums are associations and their ranking is unstable. Whole-word skill matching misses implied skills, and two notebooks differ for a few skills because their alias lists differ (Machine Learning +21% against +14%). Google Trends is a sample of attention, and the raw data cannot be shared.

Table 9. Objective-to-evidence traceability matrix.

Objective / KPI	Analysis / method	Result / metric	Figure / artifact	Recommendation
O1 Estimate pay	Section 4	Test R^2 0.767; error ₹3.8 L vs ₹5.6 L	Table 5; Fig. 3	R1, R2
O2 Range that holds	Section 4	83.3% at an 80% target; 59% to 97% by role	Fig. 3	R2
O3 Skills and pay	Section 5.1	17 of 170 clear; rank agreement 0.28	Fig. 4	R3
O4 Role from text	Section 5.1	Accuracy 0.897; macro-F1 0.881	textmining.json	R4
O5 Who states pay	Section 5.1	F1 0.518; AUC 0.871; 76.5% from never-disclosers	Table 6	R5
O6 Demand and interest	Section 5.2, 6.1	3 rising, 12 declining; no method beats naive	Fig. 5	R3, R6

8.2 Project tracking and individual work

Work was tracked through the repository's issue tracker at https://github.com/anusanth26/Career-Market-Intelligence/issues (27 issues, all closed) and through pull requests and commits. The project board is at https://github.com/users/anusanth26/projects/1. Individual marks can vary with participation and the explanation each member gives. Table 10 lists each member's main work with a link to it.

Table 10. Individual contributions.

Team member / register no.	Assigned work	Evidence link / artifact
V Kireeti, CB.SC.U4CSE23057	Text mining: skill extraction, LDA and NMF topics, K-means clusters, Louvain skill bundles, skill pay associations (least squares, ridge, double machine learning, false-discovery control, placebo and stability tests), role fit, pay disclosure and career switch	Text mining notebook
R Anusanth, CB.SC.U4CSE23235	Time series: Google Trends collection in batches with anchor rescaling, trend labels with Mann-Kendall test, naive, smoothing, ARIMA and Holt-Winters forecasts with rolling-origin backtest, STL seasonality, conformal intervals and break test	Time series notebook
Mithresh M, CB.SC.U4CSE23039	Combined view: employer demand from the postings joined with search interest and pay association, quadrant rule and skill gap table. Salary modelling, second part: stacking, tuned XGBoost with randomised search, held-out test evaluation (R^2, mean error, share within 30%), permutation importance, conformal 80% range and coverage by role	Combined insight notebook, Career market intelligence notebook
Ravula Akshith, CB.SC.U4CSE23648	Salary modelling, first part: baselines (global and role median), Ridge and random forest, employer-grouped 5-fold cross-validation and the model comparison table	Career market intelligence notebook
Rishinath S Kurup, CB.SC.U4CSE23741	Data preparation and features: merging runs, removing repeated postings, salary screens, role mapping, experience, seniority, city and skill features, text SVD and the leakage audit; also the dashboard application and the Careers page	Frontend, Careers page, Career market intelligence notebook

Technical detail of the dashboard and Careers work (Rishinath). The dashboard is a Next.js 16 application on React 19 and TypeScript, using Observable Plot and d3 for charts and TanStack Table for the dataset view. A shared filter state (role, city, experience, work mode) is kept in the web address, so every page reads the same selection. The pages read JSON files that the notebooks export, from the career_market_intelligence notebook (salary.json, dataset.json, calculator.json) and the text-mining notebook (switch.json). The Careers page runs the skill search in the browser: it adds, one at a time, the skill that most raises the target role's score, limited to skills at least 10% of the target role's postings ask for. The finished application was checked with a scripted browser session of about 40 steps.

8.3 AI use and originality declarations

We used an AI assistant as a tool during this project. It helped write and debug notebook code, build the dashboard, draw figures and draft and format this report.

The team chose the problem, the roles and skills to study and the data sources. We collected the Naukri postings and Google Trends data, decided which models and tests to run, and checked the results against the notebooks. All members have read the report and take responsibility for what it says.

9. Conclusion, Recommendations and Roadmap

The objectives were met as shown in Table 9. The salary model beat the role median on unseen employers (O1), its 80% range held overall but not for the smallest roles (O2), most skill premiums did not survive checking (O3), the description identifies the role and predicts disclosure (O4, O5), and search interest differs from demand (O6).

Table 11. Prioritised recommendations.

No.	Recommendation	Expected KPI effect	Stakeholder	Feasibility
R1	Negotiate from a range and compare with similar postings, not with one number	Offer placed against the 80% range and similar postings	Job seekers	High; in the dashboard now
R2	Read the range with its role-level coverage; widen it for Software Engineer, Data Scientist and Product Manager	Fewer under-covered benchmarks	Recruiters, HR	High
R3	Treat Python and SQL as entry requirements and add Machine Learning or cloud skills for data roles; do not choose skills by search interest	Skill plans linked to demand and pay evidence	Students, trainers	Medium; premiums are weak
R4	Put the role's own content in the first lines of a description	Keeps role signal; first 500 characters score lower	Recruiters	High
R5	State a salary range in the posting, and report how many postings stand behind any benchmark	Higher disclosure share	Recruiters, analysts	Medium
R6	Collect postings repeatedly before relying on skill pay figures	Stable premiums	Project team	Medium

Roadmap. Next month: show range reliability by role beside every estimate, with the benchmark sample size. In three to six months: repeat the collection monthly, re-estimate skill premiums and add a second board. In six to twelve months: extract skills with a trained model, model both ends of the range, and test the skill advice against real career moves. Success checks are coverage within five points of 80% for each role with enough postings and a premium rank agreement above 0.5 on held-out employers.

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Appendix A: Data validation records

Validation records: 14,085 unique job ids; 1,709 postings (12.1%) with a yearly salary range; 11,972 marked not disclosed, 352 unpaid and 52 with a monthly amount; 790 repeated descriptions removed from the text corpus; 32 implausible and 9 repeated salary rows removed. Raw files are held locally and are not included.

Appendix B: Additional analysis

Table 12. Supplementary results.

Check	Result
Salary model without text / with 50-component text summary	CV R^2 0.673 / 0.696
Text limited to first 500 characters	CV R^2 0.677
Pay amounts removed from the text	CV R^2 0.694
Learning curve, training share 20% to 100%	CV R^2 about 0.62 rising to 0.70
Role fit, macro-F1 by split ratio 80/20, 70/30, 60/40	0.88 to 0.89, stable
Disclosure, ROC-AUC by split ratio 80/20 to 50/50	0.859 to 0.866
Pay-premium stability, rank agreement on held-out companies	About 0.28 at 80/20, 70/30 and 60/40
Search-interest forecasts, share of 80% conformal intervals that held	Naive 86.4%, smoothing 87.5%, ARIMA 89.5%, Holt-Winters 78.6%

Appendix C: Implementation, configuration and dashboard guide

Install with `pip install -r requirements.txt` for the notebooks and `cd frontend && npm install` for the dashboard. Run the four notebooks in order, then `npm run dev` in the frontend folder and open http://localhost:3100. Set role, city, experience and work mode in the filter strip; open Pay for the estimator and offer check, Skills for the demand map and skill lookup, Careers for the career switch planner, and Data and method for the dataset. Role rules are in config/roles.json and tracked skills in config/review2_skills.json.

Appendix D: Tracking evidence, contribution records, declaration and similarity report

Issue tracker: https://github.com/anusanth26/Career-Market-Intelligence/issues. Project board: https://github.com/users/anusanth26/projects/1. Commit history is available from the repository.